CSA3714 Software Testing

CO1 - AT2 Short Answers

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2) Standardized Process: Replaces ad-hoc testing with a structured lifecycle to ensure systemic validation.

Impact & Benefits: Ensures high transaction reliability and prevents financial calculation errors. It also helps maintain regulatory security standards.

7) Origin of Defects: They originate in early SDLC phases vague tax or salary rules cause developers to code incorrect logic.

Prevention: Clear, unambiguous requirements and acceptance criteria give testers precise benchmarks.

8) Classification: Boundary value defect, caused by software failing at operational limit in conditional logic.

Prevention: BVA tests input at the, just below, just above boundary line to catch edge case crashes.

11) Defect clustering: Most Defects are concentrated in a small number of modules.

Testing Efficiency: Testing team use historical data to prioritize high risk modules, maximizing defect detection.

14) Role of a Tester: Acts as a quality bridge by translating developer logic into test scenarios and sharing detailed defect logs.

Early Involvement: Prevention integration gaps by giving testers a clear understanding of system architecture before coding starts.

18) Cross Browser Strategy: Design a strategy utilizing automated cross browser testing (Chrome, Firefox, Safari)

Handling Issues: Resolves layout, script, execution, user interface discrepancies to ensure a consistent customer experience.

19) Validation Gaps: Caused by a lack of rigorous input validation, unhandled exceptions, missing data type.

Improving Reliability: Proper validation ensures data integrity, preventing unpredictable outputs and improving overall system.

25) Redesigning approach: Replace ad-hoc methods with a structured testing framework featuring.

Enhancing Reliability: Structured testing prevents recurring transaction faults and guarantees high operational accuracy.

26) Including Edge Cases: Implement Boundary Value Analysis (BVA) to test limits and extreme values rather than relying on normal inputs.

Improving Defect Detection: Uncovers hidden logic faults overflow vulnerabilities and off-by-one errors at operational thresholds.

30) Redesigning Strategy: "Shift Left" by incorporating test planning, requirement reviews and static testing into the earliest phases of the SDLC.

Improving Efficiency: catching Defects early reduces the exponential costs and effort required to fix the bugs discovered late in production.